

UX/UI Course Capstone · Pre-Design Phase

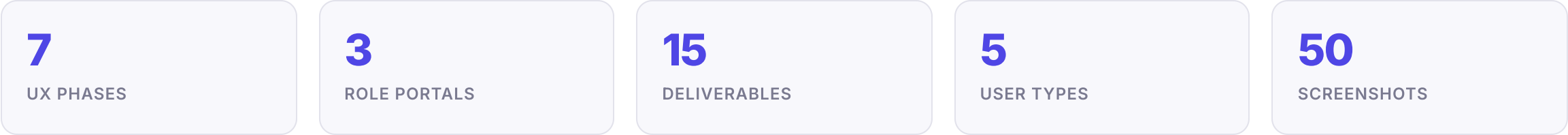
Travelex

T&E Platform

PROCESS DOCUMENTATION

A structured, AI-assisted UX methodology from problem definition to implementation-ready handoff.

Project at a Glance



ATTRIBUTE	DETAIL
Project Type	UX/UI Capstone — Pre-Design Phase
User Types	Employee · Manager · Finance · Admin · Executive/CFO
Portals Implemented	Employee · Manager · Finance (Admin + Executive consolidated)
Key Frameworks	Kano Model · MoSCoW · JTBD · RBAC · ERD · Journey Mapping
Handoff	Full-stack Travelex app + NotebookLM knowledge base

The Business Problem

T&E management runs on email threads, paper receipts, and spreadsheets at most companies. The tension is structural — four groups, four different definitions of "done."

WHO SUFFERS

- **Employees** — lose receipts, no visibility on where their reimbursement is
- **Managers** — approval bottlenecks, no policy context, can't delegate
- **Finance** — manual GL coding, violations caught after the fact
- **Executives** — spend data arrives weeks late, no proactive alerts

WHAT TRAVELEX SOLVES

- Pre-trip travel approval with budget tracking
- Expense capture with AI-assisted GL coding
- Multi-tier approval workflow with policy flags at submission time
- Real-time spend analytics for finance and management

The Six-Step Methodology

Each phase was treated as a gate — outputs had to be sufficient before proceeding. Iteration happened *within* each step, not across them.

#	PHASE	KEY OUTPUT
1	Problem Definition	Bounded problem statement, scope decisions, success criteria
2	Research	Market research report, user sentiment synthesis
3	Personas & User Stories	5 personas with JTBD framing and user stories
4	Journey Mapping	3 end-to-end flows, cross-role handoff diagram
5	Feature Prioritization	Kano + MoSCoW matrix, MVP scope locked
6	Information Architecture	Sitemap, user action tree, wireframe prompts

Every deliverable was drafted by AI, critiqued by the designer, and iterated — almost nothing was accepted on first pass.

01

Problem Definition

Framing the T&E problem, setting scope boundaries, and defining measurable success criteria before any requirements were written.

Problem Definition

IN SCOPE

- Travel pre-approval requests
- Expense capture and submission
- Manager approval workflow
- Finance GL coding and policy enforcement
- Spend reporting and analytics

EXPLICITLY OUT OF SCOPE (MVP)

- In-app travel booking engine
- Corporate card auto-reconciliation
- ERP / accounting system integration
- Procurement and vendor payments

Success criteria defined:

- Reimbursement cycle time reduction
- Policy compliance rate improvement
- Finance month-end close time reduction

Excluding a booking engine was the single most important scope decision — it would have doubled the technical surface area and shifted the product toward a travel marketplace model.

02

Research

Market analysis of the T&E SaaS landscape and synthesis of user sentiment from public sources.

Research Findings

WHAT INCUMBENTS DO WELL

- Approval workflow routing
- OCR receipt capture
- Corporate card integration
- Policy rule engines

WHERE THEY FALL SHORT

- Employee submission experience — an afterthought
- Over-complex admin configuration
- Weak real-time analytics for executives
- Policy feedback at rejection, not submission

Design differentiator identified:

The employee-side experience is consistently rated lowest in reviews. Most tools are optimised for Finance — the person submitting the expense is a secondary consideration. Positioning Travelex around a **fast, low-friction submission flow with policy guidance at point of entry** became the core design principle carried forward.

03

Personas & User Stories

Five user types grounded in real frustration patterns, each with JTBD framing and representative user stories.

Persona — Employee

Alice · Engineering

Submits 8–12 expenses per month · High travel frequency

Primary goal	Submit quickly, get reimbursed without friction
Key frustration	Policy rules are unclear at submission; rejection arrives days later with no explanation
JTBD	<i>Hire the platform to make submitting expenses so fast it becomes a non-event</i>

User stories:

- As an employee, I want to photograph a receipt and have the amount pre-filled, so I can submit in under 60 seconds.
- As an employee, I want to see where my report is in the approval chain, so I can follow up with the right person.
- As an employee, I want a clear rejection reason, so I can resubmit without going back and forth by email.

Persona — Manager

Mike · Engineering Manager

Reviews 15–25 expense reports per month · Frequently travelling

Primary goal	Approve accurately and quickly without becoming a bottleneck
Key frustration	Approval requests arrive through email with no policy context
JTBD	<i>Hire the platform to handle approval overhead so I can focus on my team</i>

User stories:

- As a manager, I want a policy violation summary before I approve, so I can decide without reading every line item.
- As a manager, I want to delegate approval authority when I am out of office, so reimbursements are not blocked.
- As a manager, I want a daily digest of pending approvals — not one email per request.

Persona — Finance

Sarah · Finance Manager

Owns GL coding, policy enforcement, and month-end close

Primary goal	Close the books accurately and on time; produce audit-ready records
Key frustration	Approved reports arrive requiring manual GL coding; violations caught too late
JTBD	<i>Hire the platform to eliminate manual reconciliation and give me confidence in compliance</i>

User stories:

- As a finance team member, I want AI-suggested GL codes pre-populated on approved reports, so I can confirm rather than assign from scratch.
- As a finance team member, I want a dedicated coding queue for batch processing of transactions.
- As a finance team member, I want spend-by-category reports filterable by department and date range.

Personas — Admin & Executive

ADMINISTRATOR

IT / Operations · Configures the platform

- **Goal:** Configure policies and users without engineering support
- **Frustration:** Policy changes require IT tickets in legacy tools
- **JTBD:** *Hire the platform to run itself with minimal maintenance*

Implementation note: Admin functions merged into the Finance role in Travelex.

EXECUTIVE / CFO

C-Suite · Needs real-time spend visibility

- **Goal:** Real-time spend visibility; identify budget risks early
- **Frustration:** Spend data arrives at month-end, no proactive alerts
- **JTBD:** *Same visibility into T&E spend as into revenue*

Implementation note: Executive analytics live under Manager → Team in Travelex.

04

Journey Mapping

End-to-end flows mapped across roles — with the handoffs between them made explicit.

Journey Mapping — Key Insight

Three flows mapped: **Employee submission → Manager review → Finance processing.**

The most valuable output was making cross-role handoffs explicit:

HANDOFF POINT	FROM → TO	WHAT PASSES	FAILURE MODE
Submission	Employee → Manager	Report + line items + policy flags	Manager has no context to decide quickly
Approval	Manager → Finance	Approved report + review note	Finance still needs to manually assign GL codes
Coding	Finance → Payment	GL - coded report	Report sits in queue; employee has no visibility
Rejection	Manager → Employee	Review note	No explanation → incorrect resubmission
Escalation	Manager → Finance	High - value report	No threshold - based routing

*The moment a manager approves a report and it enters the finance queue is **completely invisible to the employee** — this is why "where is my reimbursement?" is the most common T&E complaint. This surfaced a core requirement: a timestamped audit trail accessible to employees at all times.*

05

Feature Prioritization

Dual-framework approach: Kano Model for user value classification, MoSCoW for release scoping.

Kano + MoSCoW Matrix

FEATURE	KANO	MOSCOW	RELEASE
Receipt capture (OCR photo-to-form)	Basic	Must Have	MVP
Multi-tier approval workflow	Basic	Must Have	MVP
Policy rule engine (per-category limits)	Basic	Must Have	MVP
Reimbursement status tracking	Performance	Must Have	MVP
Mobile-first submission flow	Performance	Must Have	MVP
Real-time spend dashboard	Performance	Should Have	MVP
AI-suggested GL code assignment	Excitement	Should Have	MVP
Delegation of approval authority	Performance	Should Have	MVP
Bulk CSV export	Basic	Must Have	MVP
Anomaly / duplicate receipt detection	Excitement	Could Have	Release 2
ERP integration	Basic	Won't Have	Release 2
In-app travel booking engine	Indifferent	Won't Have	Release 3

06

Information Architecture

Sitemap, RBAC-shaped user action tree, and wireframe prompts — the handoff layer between discovery and visual design.

Portal Navigation — Implemented

RBAC principles were used as an IA-shaping tool: mapping create/read/update/delete permissions per role per entity determined what appeared in each portal's navigation.

PORTAL	NAVIGATION
Employee	Dashboard · Expenses · Travel · Settings
Manager	Dashboard · Approvals · Expenses · Travel · Team (+ Executive analytics) · Analytics · Settings
Finance	Dashboard · Policies · Expenses · Travel · Coding Queue · Analytics · Export · Settings

Key IA decision: The **GL Coding Queue** is a dedicated sequential interface — not embedded in the expense list. Finance team members process transactions in batches; context-switching between review and coding is a known source of errors. This decision came directly from the Finance journey map.

Data Model — ERD Entities

The ERD was produced during the pre-design phase — not delegated to engineering — because data model decisions have direct UI consequences.

CORE ENTITIES (IMPLEMENTED)

- **User** — role: employee / manager / finance
- **ExpenseReport** — status: draft → submitted → under_review → approved → rejected → paid
- **ExpenseItem** — category, amount, receipt URL, policy violation flag
- **TravelRequest** — destination, dates, budget, status, reviewer

SUPPORTING ENTITIES

- **PolicyRule** — per-category spending limits and receipt thresholds
- **Notification** — typed, user-targeted alerts with read state

Note: Pre-design ERD anticipated a separate *Approval* entity. Implementation embedded approval state as fields on ExpenseReport and TravelRequest — simpler and sufficient for the access patterns needed.

07

Visual Design Phase

Modern design system applied to the implemented platform —
frosted glass cards, indigo brand, animated data visualization.

Design System — Tokens & Aesthetic

DESIGN TOKENS (MESH-INSPIRED)

- **Brand:** Periwinkle `#5C66F5` (`--brand-600: 92 102 245`) — updated from indigo after Mesh competitive analysis
- **Light surface-0:** lavender-grey canvas `#F5F6FA`
- **Light surface-1:** white cards `#FFFFFF`
- **Dark surface-0:** deep navy `#12121A`
- **Sidebar:** always-dark `#1A1A28` — fixed in both themes

MESH-INSPIRED AESTHETIC

- **.card:** `bg-white/80 backdrop-blur-xl border border-black/7 rounded-[14px] shadow`
- **Sidebar:** dedicated CSS var tokens — dark panel independent of page theme
- **Buttons:** pill-shaped `rounded-full` matching Mesh CTAs
- **Font:** Inter — SF Pro-inspired weight + tracking
- **Sidebar subtitle:** "EXPENSE INTELLIGENCE"

Dashboard — Bento KPI Grid

LAYOUT & CHARTS

- Bento asymmetric layout — Total Spend spans 2 cols
- KPI cards with accent strip + mini sparkline + delta % badge
- Period toggle 7d / 30d / 90d filters trend chart
- Gradient area chart with animated entry
- Illustrated empty states with CTA
- Hover arrow animations on expense/travel rows

NEW COMPONENTS BUILT

- **Sparkline** — mini AreaChart, no axes, gradient fill
- **Delta** — green/red trend badge with icon
- **KpiCard** — accent strip + sparkline + delta
- **ChartTooltip** — custom dark/light tooltip
- **PeriodToggle** — 7d/30d/90d pill buttons
- **EmptyState** — icon + message + CTA

Sidebar & Shell Upgrades

SIDEBAR (MESH-STYLE)

- **Always-dark panel** — fixed #1A1A28 independent of page theme
- Periwinkle active indicator + icon tint on selected item
- Collapsible to 60px icon-only mode (localStorage persisted)
- Live pending count badges — dot on icon when collapsed, pill when expanded
- "EXPENSE INTELLIGENCE" branding (small-caps, letter-spaced)

APPLICATION SHELL

- **AI Chat Widget** — Claude API, Sparkles button in header
- **Notifications panel** — real-time unread badge, mark-read, typed alerts
- **Dark mode toggle** — Sun/Moon, localStorage persisted, full CSS var swap
- **Role switcher** — header dropdown for demo account switching
- **Global search** — cross-record query bar

Analytics — Upgraded Data Visualization

CHART	BEFORE	AFTER
Monthly Trend	LineChart (no fill)	AreaChart + gradient fill + animated entry
Category Split	PieChart (flat)	Donut chart + legend list
Department	Horizontal bar	Gradient horizontal bar (indigo to lighter indigo)
All charts	Basic tooltip	Custom dark/light styled tooltip
Trend chart	Fixed time window	Period toggle (7d / 30d / 90d)

Additional details: rounded bar tops on all bar charts (`radius={[4,4,0,0]}`); donut `innerRadius` added to PieChart; linearGradient left-to-right on department bars; KPI cards upgraded with colored icons + delta badges.

Modern UX — Research-Driven Upgrades

Before implementing, competitive research was conducted across Ramp, Brex, Navan, Expensify, Airbase, and Spendesk — identifying 10 modern patterns to implement.

INSPIRATION	PATTERN IMPLEMENTED
Ramp / Brex	Bento asymmetric KPI grid
Navan	Live pending badge counts on sidebar
Brex	Sparklines + delta % on KPI cards
Ramp / Airbase	Period toggle on analytics
Spendesk	Gradient fills under area charts
Expensify	Illustrated empty states with CTA
Mesh Payments	Always - dark sidebar · periwinkle brand · pill buttons · SVG empty - state illustrations
Ramp	Sortable column headers
Figma designs	2 - column form layout + Tips card

Deliverables Produced

DELIVERABLE	PURPOSE
Problem statement	Scope boundary, success criteria — north-star reference
Market research report	Competitive context; ingested into NotebookLM
Five personas + user stories	Foundation for all downstream design decisions
Journey maps (3 flows)	Cross-role handoff visibility; surfaced status-tracking requirement
Kano + MoSCoW matrix	MVP scope — defensible to design and engineering audiences
Sitemap + user action tree	Navigation structure per role; source of truth for RBAC
ERD	Backend data model; produced pre-design not post
Wireframe prompts (3 portals)	Primary handoff artifact for visual design phase
Travellex — full-stack implementation	React + FastAPI + SQLite; 3 role-based portals
Consolidated export + NotebookLM corpus	Queryable design knowledge base
Visual design system + component library	Mesh-inspired periwinkle brand; always-dark sidebar; pill buttons; 8 reusable components
Application shell	AI Chat (Claude API), notifications, dark mode, role switcher, global search
50 UI screenshots (1440×900)	Full coverage: 3 roles × all pages, light + dark mode

AI-Assisted Workflow — The Iteration Loop

WHERE AI EXCELLED

- **Structural scaffolding** — first-draft frameworks and templates at full depth
- **User story generation** — grammatically and structurally correct at scale
- **Diagram syntax** — valid Mermaid / ERD notation reliably
- **Edge case surfacing** — volunteered scenarios not explicitly considered
- **Consistency enforcement** — flagged decisions that contradicted prior outputs

WHERE HUMAN JUDGMENT WAS ESSENTIAL

- **Deciding which users matter** — Admin vs. Finance sub-role required domain judgment
- **Grounding personas** — AI defaults are optimistic; real frustrations needed pushing
- **MVP scoping** — tradeoffs no framework can make automatically
- **Navigability validation** — logical grouping ≠ navigability; required UX judgment to flatten

The standard loop: seed context → AI generates draft → designer critiques → iterate → lock deliverable. Almost nothing was accepted on first pass.

Outcomes & Lessons Learned

WHAT WORKED WELL

- **Sequential phase discipline** — each phase as a gate produced more defensible later-phase outputs
- **Cross-role journey mapping** — the single most productive methodological decision
- **RBAC as IA principle** — applied before design began, not retrofitted
- **Iterative critique** — resisting first-pass acceptance separated grounded from generic

WHAT TO CHANGE NEXT TIME

- **Structured competitive benchmark** — feature comparison table vs. incumbents
- **Earlier ERD** — data model as input to feature prioritization, not output of IA
- **User testing proxies** — at least one validation round with real users in target roles

Skills reinforced: Scope definition before requirements · Complementary framework pairing (Kano + MoSCoW) · Cross-role handoff thinking · Pre-design data modelling

Travelex — Expense Intelligence Platform

Thank you

UX/UI Capstone · Pre-Design Phase Documentation

3 role-based portals · 7 phases · 15 deliverables · 50 screenshots

Built with Claude · React · FastAPI · Travelex